

From Uncertain Predictions to Uncertain Explanations with Gaussian Processes

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A limited budget for experiments

Which electrolyte recipe should we test next, and why?

What we can choose (the x independent variable)

Salt concentration and solvent composition.

What we measure (the y dependent variable)

Ionic conductivity: how readily ions move through the electrolyte.

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The same prediction can support different experimental choices.

Uncertainty matters because it changes what we should do next. Is it noise? or opportunities?

The questions behind an experiment

A useful model supports scientific reasoning beyond a point prediction.

1. **What might happen?**

Plausible response functions and uncertainty.

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3. **What should we measure next?**

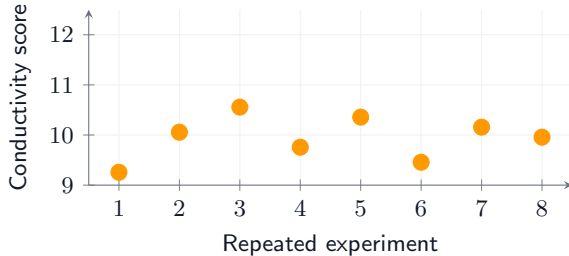
Experiment selection informed by the model and an expert.

Section 1

Uncertainty and Gaussian processes

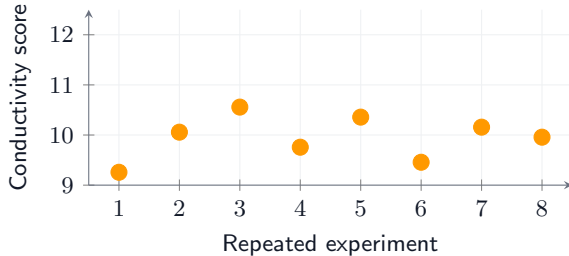
Aleatoric uncertainty

Repeating the same recipe can give different measurements.



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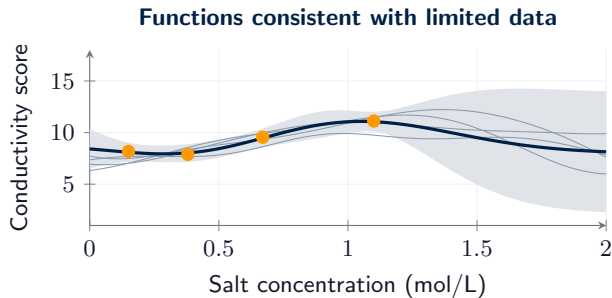
Aleatoric means variability remaining at fixed recorded inputs.

It may reflect measurement noise or unrecorded conditions.

Repetition estimates the mean more accurately; a single future measurement still varies.

Epistemic uncertainty

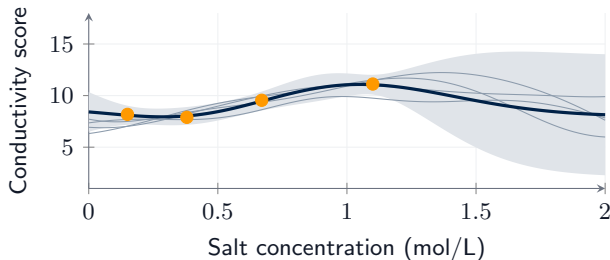
With few observations, several response functions remain plausible.



Epistemic uncertainty

With few observations, several response functions remain plausible.

Functions consistent with limited data



Epistemic means uncertainty about the underlying response function.

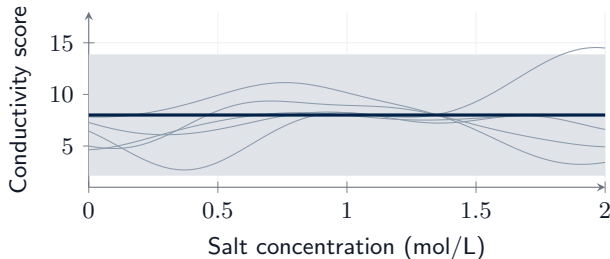
The curves agree more near observations and differ in untested regions.

An informative new experiment can reduce this uncertainty.

A distribution over functions

A Gaussian process represents many plausible curves and how they co-vary.

Before observing conductivity data



Mean function

The average curve.

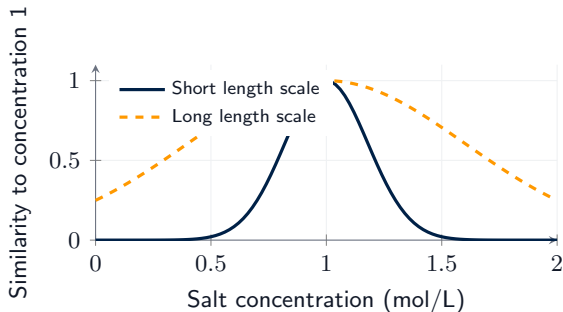
Covariance function

How strongly responses at two inputs vary together.

Nearby recipes can share information.

A kernel expresses assumptions about similarity

Measurements at similar recipes can inform one another.



Squared-exponential kernel; illustrative length scales.

A **kernel** is a covariance rule connecting pairs of inputs.

$$k(x, x') = a^2 \exp\left(-\frac{(x - x')^2}{2\ell^2}\right).$$

a : typical function variation.

ℓ : distance over which values stay similar.

What makes it a Gaussian process?

Any finite set of function values has a joint Gaussian distribution.

$$f \sim \mathcal{GP}(m, k) \iff \begin{pmatrix} f(x_1) \\ \vdots \\ f(x_n) \end{pmatrix} \sim \mathcal{N} \left(\begin{pmatrix} m(x_1) \\ \vdots \\ m(x_n) \end{pmatrix}, K \right),$$
$$K_{ij} = k(x_i, x_j).$$

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A Gaussian vector

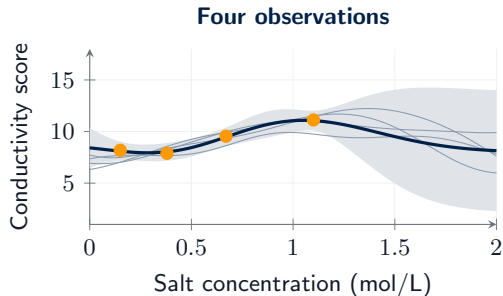
A bell-shaped joint distribution for several values, allowing them to be correlated.

A flexible response curve

The function can be nonlinear in its inputs. Gaussian describes the distribution over values.

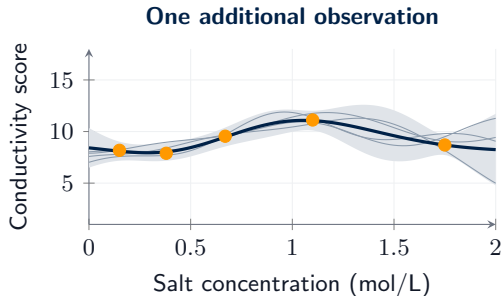
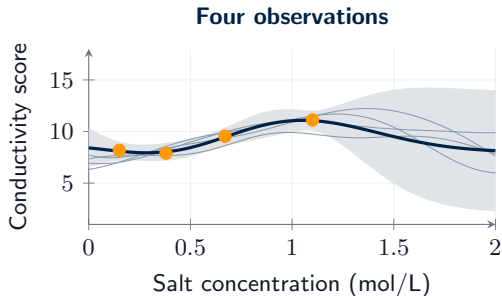
Learning updates the distribution of curves

New evidence constrains the response in the region we measure.



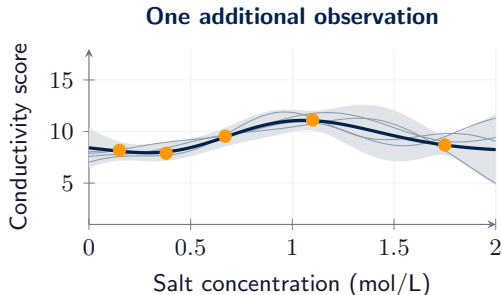
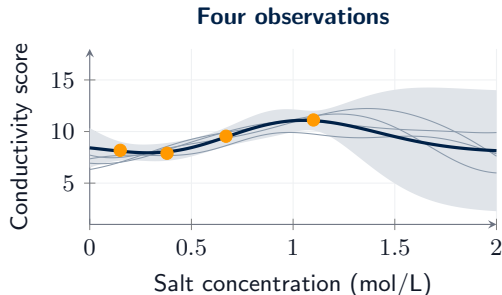
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The posterior retains several plausible curves, with less uncertainty near the new observation.

The GP regression update

The update combines similarity-weighted evidence with a reduction in uncertainty.

Assume zero prior mean and observations $y_i = f(x_i) + \epsilon_i$, with independent $\epsilon_i \sim \mathcal{N}(0, \sigma_n^2)$.

$$\underbrace{\mu_{\mathcal{D}}(x)}_{\text{updated mean}} = k_x^\top \underbrace{(K + \sigma_n^2 I)^{-1} \mathbf{y}}_{\text{weights from the data}},$$

$$\underbrace{k_{\mathcal{D}}(x, x')}_{\text{remaining covariance}} = \underbrace{k(x, x')}_{\text{prior covariance}} - \underbrace{k_x^\top (K + \sigma_n^2 I)^{-1} k_{x'}}_{\text{reduction from observing data}}.$$

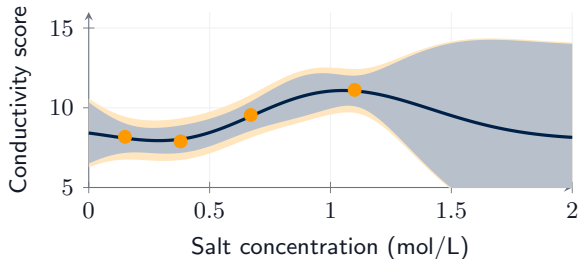
$K_{ij} = k(x_i, x_j)$; $k_x = (k(x_1, x), \dots, k(x_n, x))^\top$; I is the identity matrix.

Gaussian observation noise gives an analytic update.

Rasmussen & Williams, Gaussian Processes for Machine Learning, Ch. 2.

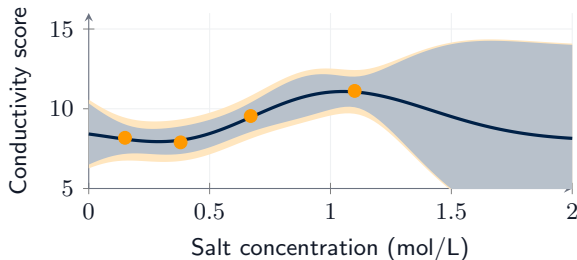
Function uncertainty and measurement noise

A future measurement includes both uncertainty about the response and observation noise.



Function uncertainty and measurement noise

A future measurement includes both uncertainty about the response and observation noise.



Blue: uncertainty about $f(x)$.

Orange: uncertainty about $y(x)$.

$$\text{Var}[y(x) \mid \mathcal{D}] = \underbrace{k_{\mathcal{D}}(x, x)}_{\text{epistemic}} + \underbrace{\sigma_n^2}_{\text{aleatoric}}.$$

This decomposition uses the stated GP and independent-noise model.

A connection to deep learning

Some neural-network priors become GP priors as their layers become very wide.

A one-hidden-layer illustration adds many randomly weighted features:

$$f_M(x) = \frac{1}{\sqrt{M}} \sum_{r=1}^M a_r \psi(w_r^\top x + b_r).$$

- ▶ ψ is the activation function; a_r, w_r, b_r are random weights and biases.
- ▶ With suitable independence, scaling and finite-moment assumptions, the joint outputs approach a Gaussian distribution as M grows.
- ▶ The resulting kernel records similarities induced by the network's random features.

GPs and neural networks can express closely connected beliefs about functions.

Lee et al., Deep Neural Networks as Gaussian Processes, ICLR 2018.

Section 2

Feature contributions

Explaining a particular recipe

We want to share a prediction's difference from a reference across its features.

Prediction to explain

This recipe has score 15, compared with a reference prediction of 8.

How much of the difference comes from salt concentration? How much from solvent composition?

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Values of feature coalitions

A **coalition** is simply a subset S of the recorded features.

Keep their values and average over the remaining features:

$$v_x(S) = \mathbb{E}_X[f(X) \mid X_S = x_S].$$

The reference distribution defines what this average means.

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These are contributions to a model prediction, relative to a chosen reference.

A two-feature prediction game

A feature's added contribution can depend on which other feature we already know.

Feature information available	Model's expected score
Neither feature: $v(\emptyset)$	8
Salt only: $v(\{1\})$	10
Solvent only: $v(\{2\})$	11
Both: $v(\{1, 2\})$	15

Salt adds $10 - 8 = 2$ when it is known first, but $15 - 11 = 4$ when solvent is already known.

Shapley values average over arrival orders

Average a feature's added contribution over every order in which features can be revealed.

Order of revealing features	Salt contribution	Solvent contribution
Salt, then solvent	$10 - 8 = 2$	$15 - 10 = 5$
Solvent, then salt	$15 - 11 = 4$	$11 - 8 = 3$
Average	$\phi_1 = 3$	$\phi_2 = 4$

$$\underbrace{3 + 4}_{\text{feature contributions}} = \underbrace{15 - 8}_{\text{prediction minus reference}}.$$

The interaction contribution of 2 is shared equally between the two features.

The general Shapley rule

The same averaging rule applies when many features interact.

For d features, let $N = \{1, \dots, d\}$ and S be a subset excluding feature i .

$$\phi_i(v) = \sum_{S \subseteq N \setminus \{i\}} \underbrace{\frac{|S|! (d - |S| - 1)!}{d!}}_{\text{fraction of orders with exactly } S \text{ before } i} \underbrace{[v(S \cup \{i\}) - v(S)]}_{\text{added contribution}}.$$

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Efficiency: contributions sum to the total change.

Null player: a feature changing no subset's value gets zero.

Symmetry: equivalent players receive equal allocations.

Linearity: allocations add when games add.

Shapley (1953); Lundberg & Lee (NeurIPS 2017). $|S|$ counts the features in S .

Section 3

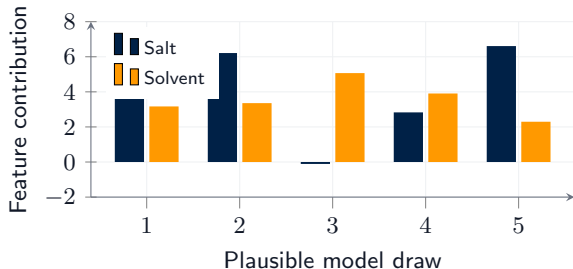
Uncertain explanations

Each plausible model has its own explanation

Uncertainty about the response function should propagate into its feature contributions.

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Uncertainty about the response function should propagate into its feature contributions.



Sample one whole function f from its posterior.

Compute that function's Shapley values.

Repeat conceptually to obtain a **distribution of explanations.**

Chau, Muandet & Sejdinovic, NeurIPS 2023. Illustrative posterior model with mean contributions 3 and 4.

Gaussian distributions of explanations

For a GP, the joint distribution of feature contributions is available analytically.

1. Averaging over missing inputs is a linear operation on f .
2. Taking weighted differences of subset values is also linear.

Collect the subset values in a vector $\mathbf{v}$; let A perform the Shapley weighted sums.

$$\mathbf{v} \sim \mathcal{N}(\mathbf{m}_v, C_v), \quad \phi = A\mathbf{v} \sim \mathcal{N}(A\mathbf{m}_v, AC_vA^\top).$$

The result gives both mean contributions and their full covariance.

Chau, Muandet & Sejdinovic, NeurIPS 2023. Gaussian game and stochastic Shapley distribution, Sec. 3.

The allocation rules hold for random explanations

A coherent allocation can retain uncertainty while obeying the same consistency rules.

$$\sum_{i=1}^d \phi_i(f, x) = f(x) - \mathbb{E}_X[f(X)].$$

This identity holds for each jointly sampled function, including its reference prediction.

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This identity holds for each jointly sampled function, including its reference prediction.

- ▶ Equivalent players receive equivalent random allocations.
- ▶ A null player receives zero contribution.
- ▶ Adding games adds their random allocations.

The uncertainty lives in a consistent joint explanation, including the baseline.

Chau, Muandet & Sejdinovic, NeurIPS 2023. Stochastic Shapley axioms, Sec. 2.

The mean explanation is only part of the answer

Explaining the average function retains the mean contribution but loses its uncertainty.

$$\mathbb{E}_f[\phi(f, x) \mid \mathcal{D}] = \phi(\mathbb{E}_f[f \mid \mathcal{D}], x).$$

The mean contribution follows from linearity. Its variance requires the dependence between subset values:

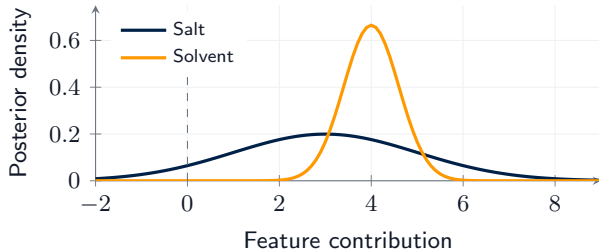
$$\text{Var}_f[\phi_i \mid \mathcal{D}] = \sum_{S, T} A_{iS} A_{iT} \text{Cov}_f[v(S), v(T) \mid \mathcal{D}].$$

Uncertainty in a contribution differs from a Shapley allocation of predictive variance.

Chau, Muandet & Sejdinovic, NeurIPS 2023. Proposition 5. Fixed reference distribution.

How certain is the direction of a contribution?

A positive average contribution can coexist with uncertainty about its sign.



Salt: mean 3, standard deviation 2.

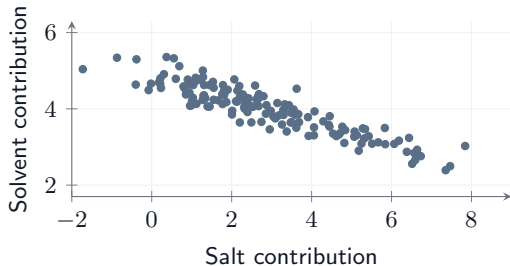
95% posterior interval: $[-0.92, 6.92]$.

Solvent: mean 4, standard deviation 0.6.

95% posterior interval: $[2.82, 5.18]$.

Feature contributions can move together

Understanding a combined explanation requires its covariance, not only separate error bars.



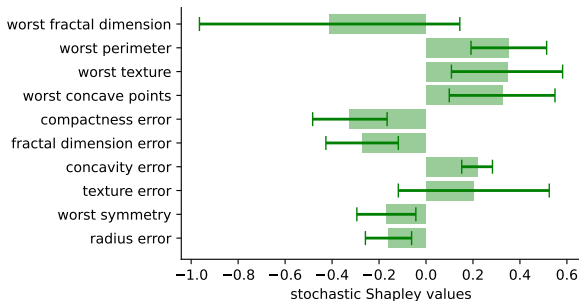
Each dot is one joint explanation.

Here, a larger salt contribution tends to accompany a smaller solvent contribution.

$$\begin{aligned}\text{Var}(\phi_1 + \phi_2) &= \text{Var}(\phi_1) + \text{Var}(\phi_2) \\ &\quad + 2 \text{Cov}(\phi_1, \phi_2).\end{aligned}$$

A case from the NeurIPS paper

Two features can have similar average contributions with very different explanation uncertainty.



Breast-cancer dataset: 569 patients, 30 numerical features.

Compare **worst fractal dimension** and **worst perimeter**.

Their mean contribution magnitudes are similar; the first is more uncertain.

Chau, Muandet & Sejdinovic, NeurIPS 2023. Fig. 2(a), BayesGP-SHAP. Bars show means with uncertainty.

Model uncertainty and calculation uncertainty

More experiments and more explanation computation address different uncertainties.

Method	What varies?	What can reduce it?
GP-SHAP	Response functions under the GP posterior.	Informative response data.
BayesSHAP	Shapley estimates for a fixed function.	More sampled feature subsets.
BayesGP-SHAP	Both sources of uncertainty.	Better data and more computation.

A precise numerical calculation can still explain an uncertain model.

Chau, Muandet & Sejdinovic, NeurIPS 2023. Sec. 3.2 and the California-housing illustration in Fig. 1.

Fast explanations for a structured GP

Separating individual effects and interactions lets us share each interaction directly.

Functional ANOVA (FANOVA) separates a function into effects of feature subsets.

For two features:

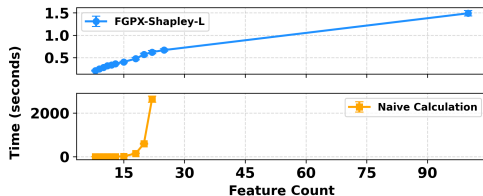
$$f(x) = f_0 + f_1(x_1) + f_2(x_2) + f_{12}(x_1, x_2).$$

For the component-based game:

$$\phi_1 = f_1 + \frac{1}{2}f_{12}, \quad \phi_2 = f_2 + \frac{1}{2}f_{12}.$$

Exact stochastic attribution with quadratic dependence on feature count, for this structured model.

Mohammadi et al., AAAI 2026. Fig. 1. Explanation computation after fitting the GP.



Section 4

Choosing and explaining experiments

What makes an experiment informative?

The best place to learn the response curve can differ from the best place to seek high performance.

Active learning

Choose observations to improve the model's knowledge.

A simple rule, with equal observation noise:

$$x_{\text{learn}} = \arg \max_x k_{\mathcal{D}}(x, x).$$

The purpose of learning determines how we use uncertainty.

Bayesian optimisation

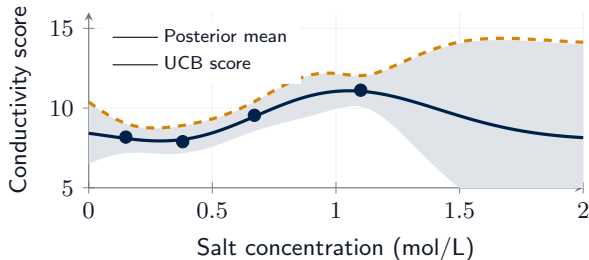
Choose observations to find a high-performing recipe using few experiments.

The ultimate target is

$$x^* = \arg \max_x f(x).$$

A score for the next experiment

An uncertain recipe can be attractive because it may outperform the best known one.



An **acquisition function** scores where to experiment next.

Upper confidence bound (UCB):

$$\alpha_t(x) = \mu_t(x) + \sqrt{\beta_t} \sigma_t(x).$$

Mean: predicted performance.

Standard deviation: exploration bonus.

Choose the recipe maximising α_t .

Srinivas et al. (ICML 2010). Illustrative GP; $\sqrt{\beta_t} = 2$ and $\sigma_t^2 = k_t(x, x)$.

Explaining performance and exploration

Separate the features supporting predicted performance from those driving exploration.

Recall the Gaussian subset predictions in GP-SHAP: m_S is their mean and $C_v(S, S)$ their variance after averaging over missing settings.

$$u(S) = m_S + \sqrt{\beta_t} s_S, \quad s_S = \sqrt{C_v(S, S)}.$$

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UCB now scores each subset. Applying the same Shapley operator A to the vectors of means $\mathbf{m}_v$ and standard deviations $\mathbf{s}_v$ gives:

$$\phi_i^{\text{UCB}} = \underbrace{(A\mathbf{m}_v)_i}_{\text{mean GP-SHAP attribution}} + \sqrt{\beta_t} \underbrace{(A\mathbf{s}_v)_i}_{\text{contribution to exploration}}.$$

A contribution to exploration differs from uncertainty in the attribution itself.

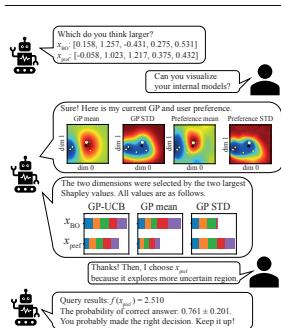
GP-SHAP (NeurIPS 2023); CoExBO (AISTATS 2024), Eq. (12). This constructs UCB for subset-averaged GP predictions; directly averaging UCB defines a different game.

CoExBO brings the scientist into the loop

An expert can compare plausible experiments using knowledge the model has not recorded.

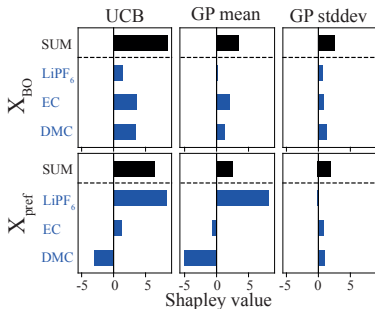
1. Propose two recipes: standard BO and preference-informed BO.
2. Explain their predicted performance and selection scores.
3. Let the expert choose between them.
4. Run the selected experiment and update both models.

Adachi et al., AISTATS 2024. Collaborative and Explainable Bayesian Optimization; Fig. 2.



Why did the optimiser suggest this recipe?

Explaining predicted performance and exploration separately helps the expert compare candidates.



Two candidates: x_{BO} and x_{pref} .

Three explanation targets:

- ▶ Selection score.
- ▶ Predictive mean.
- ▶ Predictive standard deviation.

Adachi et al., AISTATS 2024. Fig. 8. Acquisition explanations use the paper's chosen subset-value construction.

Combining objective and preference evidence

A confidently learnt preference can guide proposals; an uncertain one receives less weight.

Objective model

A GP learns conductivity from measurements.

Preference model

A second GP learns which recipes the expert favours from comparisons.

After putting both on a common scale, combine their Gaussian information:

$$\mu_{\text{mix}} = \frac{v_p \mu_f + v_f \mu_p}{v_f + v_p}.$$

μ_f, v_f : objective mean and variance.

μ_p, v_p : scaled preference mean and variance.

A large v_p reduces preference influence.

Adachi et al., AISTATS 2024. Proposition 1. Preference variance also includes an iteration-dependent decay term.

A safeguard against misleading feedback

CoExBO gradually reduces preference influence as experimental evidence accumulates.

The added variance term makes the preference weight shrink:

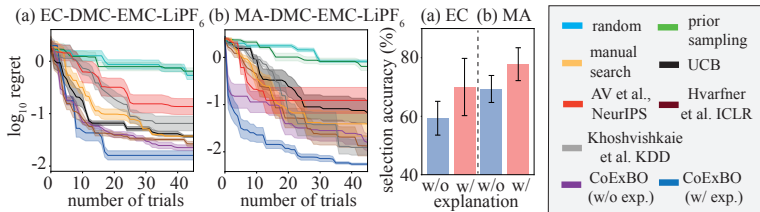
$$v_p(x) = v_{p,\text{scaled}}(x) + \gamma t^2 v_f(x), \quad w_p(x) = \frac{v_f(x)}{v_f(x) + v_p(x)} \leq \frac{1}{1 + \gamma t^2}.$$

- ▶ The preference-informed acquisition approaches the standard GP-UCB rule.
- ▶ Under the paper's assumptions, the asymptotic convergence rate is preserved.
- ▶ This safeguard does not guarantee better performance at every finite iteration.

Adachi et al., AISTATS 2024. Theorem 2 and Lemma 3, the asymptotic no-harm result. $\gamma > 0$.

Evidence from battery experts

Explanations improved expert selections in the reported electrolyte-design study.



- ▶ Four battery experts; two electrolyte systems, one familiar and one unfamiliar.
- ▶ Compare CoExBO with and without explanations, alongside optimisation baselines.
- ▶ Objectives were interpolated from existing experimental conductivity measurements.

What the next experiment might tell us depends on the current uncertainty.

Useful scientific models communicate both their predictions and the limits of their explanations.

1. **GPs represent plausible response functions.**
Their uncertainty can guide where to collect evidence.
2. **Feature contributions inherit model uncertainty.**
Their joint distribution supports more careful interpretation.
3. **Explanations can support experimental choices.**
CoExBO combines model evidence with expert comparisons.

Further Reading

Explaining the Uncertain: Stochastic Shapley Values for Gaussian Process Models

Siu Lun Chau, Krikamol Muandet, and Dino Sejdinovic.

Advances in Neural Information Processing Systems, Vol. 36 (NeurIPS 2023).

Looping in the Human: Collaborative and Explainable Bayesian Optimization

Masaki Adachi, Brady Planden, David Howey, Michael A. Osborne, Sebastian Orbell, Natalia Ares, Krikamol Muandet, and Siu Lun Chau.

27th International Conference on Artificial Intelligence and Statistics
(AISTATS 2024), *Proceedings of Machine Learning Research* 238:505–513.

Exact Shapley Attributions in Quadratic-time for FANOVA Gaussian Processes

Majid Mohammadi, Krikamol Muandet, Ilaria Tiddi, Annette Ten Teije, and Siu Lun Chau.

Proceedings of the AAAI Conference on Artificial Intelligence
(AAAI 2026), 40(29):24414–24421.



Questions and discussion

Siu Lun Chau

epiclab-sg.github.io

The reference distribution changes the question

A feature contribution needs a clear description of which recipes we average over.

Conditional reference

Average over recipes consistent with the retained feature values:

$$v_x(S) = \mathbb{E}[f(X) \mid X_S = x_S].$$

This respects dependencies between inputs in the reference data.

Neither construction by itself identifies a causal effect.

Chau, Muandet & Sejdinovic, NeurIPS 2023. Sec. 3.1. GP-SHAP studies the conditional-reference construction.

Marginal reference

Hold x_S fixed, and sample the remaining features from their marginal distribution:

$$v_x(S) = \mathbb{E}_{X_{-S}}[f(x_S, X_{-S})].$$

This can create combinations rarely seen in the reference data.

The GP update with a nonzero prior mean

We update deviations from the prior mean, then add that mean back to the prediction.

Let $\mathbf{m}_X = (m(x_1), \dots, m(x_n))^\top$ and $Q = K + \sigma_n^2 I$.

$$\mu_{\mathcal{D}}(x) = m(x) + k_x^\top Q^{-1}(\mathbf{y} - \mathbf{m}_X),$$

$$k_{\mathcal{D}}(x, x') = k(x, x') - k_x^\top Q^{-1} k_{x'}.$$

Latent function:

$$f(x) \mid \mathcal{D} \sim \mathcal{N}(\mu_{\mathcal{D}}(x), k_{\mathcal{D}}(x, x)).$$

Future noisy measurement:

$$y(x) \mid \mathcal{D} \sim \mathcal{N}(\mu_{\mathcal{D}}(x), k_{\mathcal{D}}(x, x) + \sigma_n^2).$$

Rasmussen & Williams (2006), Ch. 2. Fixed hyperparameters and Gaussian noise.

The two-feature Gaussian calculation

The toy calculation makes the uncertainty propagation an ordinary matrix multiplication.

$$\mathbf{v} = (v(\emptyset), v(\{1\}), v(\{2\}), v(\{1, 2\}))^\top,$$

$$A = \frac{1}{2} \begin{pmatrix} -1 & 1 & -1 & 1 \\ -1 & -1 & 1 & 1 \end{pmatrix}.$$

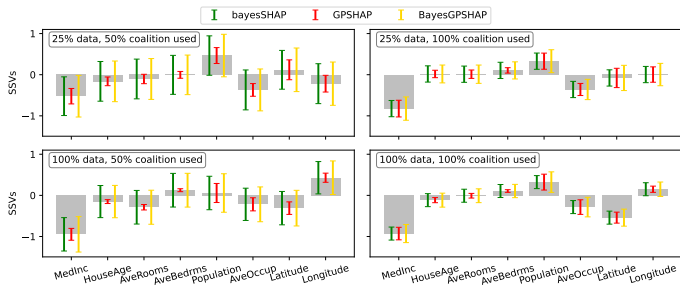
$$\mathbb{E}[\mathbf{v}] = (8, 10, 11, 15)^\top \implies \mathbb{E}[\phi] = A\mathbb{E}[\mathbf{v}] = \begin{pmatrix} 3 \\ 4 \end{pmatrix}.$$

$$C_\phi = AC_vA^\top = \begin{pmatrix} 4 & -1.08 \\ -1.08 & 0.36 \end{pmatrix} \quad \text{in the illustrative posterior model.}$$

The off-diagonal entry retains the dependence between the two explanations.

Data uncertainty and calculation uncertainty

The paper varies the training data and the number of feature subsets used for explanation.



Chau, Muandet & Sejdinovic, NeurIPS 2023. Fig. 1. California-housing dataset; reported 95% credible intervals.

Predicting explanations at new inputs

The same GP structure can share information between explanations of different recipes.

The full induced covariance connects feature i at recipe x with feature j at recipe x' :

$$\kappa_{ij}(x, x') = \text{Cov}_f(\phi_i(f, x), \phi_j(f, x')).$$

- ▶ Applying the Shapley operator to a GP prior induces a vector-valued GP prior over explanations.
- ▶ Previously computed explanations can become training observations for that GP.
- ▶ We can predict explanations at new inputs, with posterior uncertainty.

Predicting an explanation introduces a further learning problem.

Chau, Muandet & Sejdinovic, NeurIPS 2023. The Shapley prior and predictive explanations, Sec. 4.

FANOVA interaction allocations

A structured representation gives a direct allocation rule for interactions of any order.

For the component-based game,

$$f(x) = \sum_{T \subseteq N} f_T(x_T), \quad v_x(S) = \sum_{T \subseteq S} f_T(x_T),$$

$$\phi_i(f, x) = \sum_{T \ni i} \frac{f_T(x_T)}{|T|}.$$

- ▶ A two-feature interaction is shared in halves; a three-feature interaction in thirds.
- ▶ The components remain random under the GP posterior.
- ▶ Recursions aggregate kernel terms without enumerating subsets.

Mohammadi et al., AAAI 2026. Local stochastic allocation. Global sensitivity allocations address a different question.

The acquisition explanation has its own game

The function we explain and the way we average missing inputs determine the explanation.

For $\alpha(x) = \mu(x) + \sqrt{\beta} \sigma(x)$, direct conditional SHAP would use

$$v_{\text{direct}}(S) = \mathbb{E}_X[\alpha(X) \mid X_S = x_S].$$

The CoExBO paper introduces a tractable surrogate subset value:

$$v_{\text{surrogate}}(S) = \mathbb{E}_X[\mu(X) \mid X_S = x_S] + \sqrt{\beta} \sqrt{\mathbb{E}_X[k_{\mathcal{D}}(X, X) \mid X_S = x_S]}.$$

This is an upper bound on directly averaging UCB. The kernel construction in Eq. (12) instead uses the variance of a subset-averaged GP prediction, as on slide 28.

Explaining a selection score is distinct from uncertainty in an explanation of f .

Adachi et al., AISTATS 2024. Sec. 3.3, subset-value construction and its kernel-based estimate.

The preference fusion and its asymptotic safeguard

Confidence weighting and deliberate preference decay have different roles.

After scaling the preference score to the objective's scale:

$$v_p(x) = v_{p,\text{scaled}}(x) + \gamma t^2 v_f(x), \quad v_{\text{mix}}(x) = \frac{v_f(x)v_p(x)}{v_f(x) + v_p(x)},$$
$$\mu_{\text{mix}}(x) = \frac{v_p(x)\mu_f(x) + v_f(x)\mu_p(x)}{v_f(x) + v_p(x)}, \quad \alpha_{\text{mix}}(x) = \mu_{\text{mix}}(x) + \sqrt{\beta_t v_{\text{mix}}(x)}.$$

No-harm scope: under the paper's finite-domain UCB confidence and regret assumptions, preference decay preserves the asymptotic convergence rate.

Adachi et al., AISTATS 2024. Proposition 1, Theorem 2 and Lemma 3. $\gamma > 0$.

Selected references

Main research

Chau, Muandet & Sejdinovic (2023). *Explaining the Uncertain: Stochastic Shapley Values for Gaussian Process Models*. NeurIPS.

Adachi et al. (2024). *Looping in the Human: Collaborative and Explainable Bayesian Optimization*. AISTATS.

Mohammadi et al. (2026). *Exact Shapley Attributions in Quadratic-time for FANOVA Gaussian Processes*. AAAI.

Foundations

Rasmussen & Williams (2006). *Gaussian Processes for Machine Learning*.

Shapley (1953). *A Value for n -Person Games*.

Lundberg & Lee (2017). *A Unified Approach to Interpreting Model Predictions*. NeurIPS.

Lee et al. (2018). *Deep Neural Networks as Gaussian Processes*. ICLR.

Srinivas et al. (2010). *Gaussian Process Optimization in the Bandit Setting*. ICML.